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Prevalence of and secular trends in diagnosed diabetes in Italy: 1980–2013

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Abstract *Background and aims:* The aim of this research was to examine the prevalence of diabetes in Italy over a 34-year period.
Methods and results: Self-reported diabetes was assessed in eight health interview surveys of representative samples of Italian population aged 20 years and over. Crude and standardised prevalence were calculated by age, sex, educational level and area of residence. Logistic models were fitted to calculate the contribution of age and BMI to the trend in prevalence. In 2013 nearly 3.4 million Italians had a diagnosis of diabetes, more than twice as many as in 1980. The crude prevalence of diabetes in men rose from 3.3% in 1980 to 7.1% in 2013 (+115%), and from 4.7% to 6.8% in women (+45%). The prevalence was almost stable during the eighties, and started to rise from the beginning of the nineties. One third of the increase in men and two thirds in women is due to the ageing of the population, since the age-standardised prevalence increased by 79% in men and 14% in women. The prevalence of overweight and obesity increased less steeply than diabetes, and their contribution to the trend in diabetes is less relevant than age. Prevalence rose more in the elderly, in low-educated men, and in high-educated women.
Conclusion: Given that the ageing population plays a considerable role in explaining the trend, and that the number of people in the oldest age groups will continue to grow, the rise in the number of individuals with diabetes will represent a severe challenge for the national health system.
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Introduction

The number of people with diabetes is increasing worldwide, especially in developing countries, and will continue to rise in the next decades; however, there are large inter-country differences both in the prevalence of diabetes and in its growth dynamics [1–4]. According to the Italian Institute of Statistics (ISTAT) the crude prevalence of diabetes in Italy in 2011 was 4.9%. In 2011 there were 800,000 more people with diabetes than in 2000 and age-adjusted prevalence increased by 18% in the same period. There is a North-South gradient in the prevalence of diabetes, the Southern regions being more disadvantaged. Furthermore, diabetes prevalence is higher among people with a lower level of education, in particular among women [5,6]. The time trend in prevalence of a disease is related to the time trends in incidence and survival. The incidence of type 2 diabetes, which accounts for more than 90% of cases, is

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related, other than to increasing age, to well-known risk factors such as adiposity, physical inactivity, and cigarette smoking [3], while increasing survival of people with diabetes is related to improvements in the management of the disease [1,4]. Data on the trend in prevalence report different patterns in different countries [7]. In the last 20 years, in the US, diabetes prevalence has increased by a similar magnitude across all age groups [2], while, in Canada, prevalence has increased to a greater extent among the younger population [4]; unfortunately, similar data encompassing a long time trend are scarce in Europe [1,3], and, as for Italy, the time trend in prevalence has not been studied extensively or has only involved limited geographical areas [8,9]. Using data from the Italian Health Interview Surveys, we describe the secular trend in the prevalence of self-reported diabetes in representative samples of the Italian population over a 34-year period.

Methods

Study population

The study population comprised nationally representative samples of non-institutionalised people aged≥20 years who took part in the Italian Health Interview Surveys (HIS) conducted by the National Institute of Statistics (ISTAT) in the years 1980, 1983, 1987, 1990, 1994, 2000, 2005, 2013. Since 1994 the HIS has been part of a Multipurpose Surveys System to represent the Italian non-institutionalised population and families on different topics of everyday life. The HIS uses a complex multistage stratified sampling design with appropriate design variables and a system of weights, aimed to provide region-specific estimates of the prevalence of health-related behaviours, chronic diseases and health service utilisation [10]. The surveys are based on a structured questionnaire with standardised questions and items on health status; the questions regarding diabetes and other variables used in this study did not change in any of the surveys.

Measurements

Diabetes status was assessed by asking respondents aged 20 years and over "have you ever been told you have diabetes?"; the questionnaire did not distinguish between type 1 and type 2 diabetes, or gestational diabetes. Body Mass Index (BMI) was calculated as weight (kilograms) divided by height (metres) squared based on self-reported weight and height. People were considered as overweight if their BMI was between 25.0 and 29.9, or obese if their BMI was ≥30.0. Data on weight and height were not collected in the 1980 and 1987 surveys.

Age was analysed as a categorical variable, grouped into three classes (20–49, 50–64, and ≥65).

Educational level was classified according to three levels: high (university/high school, i.e. ≥13 years of education), medium (middle school, i.e. up to 12 years of education), and low (primary school/no formal education, i.e. <8 years of education).

Area of residence was grouped into three classes: North, Centre, South.

Statistical analysis

Prevalence estimates were calculated using survey-specific weighting factors provided by the National Institute of Statistics (ISTAT) [10]. We calculated crude and age-standardised prevalence of diabetes and overweight/obesity with 95% confidence intervals, stratified by year, sex and geographical area of residence. Age and sex distribution of the Italian population in 2001 (national census) was used as the standard population.

Multivariate logistic regression models were fitted to evaluate the independent contribution of year of survey, age and BMI on diabetes, using 1983 as a reference. We first built the base model for prevalence as a function of survey year; next, we added age categorised in 5-year intervals (20–25,… ≥85), then, in a third model, we added BMI categorised as normal, overweight, and obesity. A further model including educational level was tested, but was excluded from the final tables as the results did not change. We calculated Odds Ratios (OR) and 95% confidence intervals. The log-likelihood ratio test was used to compare the goodness of fit of the models. All analyses were conducted using SAS System, Version 9.3.

Results

Trends in characteristics of the samples between 1980 and 2013 are shown in Table 1. Across the eight surveys, the percentage of people aged 65 and over, of those with a high educational level and those who were overweight/obese increased, while the percentage of participants from the different areas of Italy remained almost constant.

Between 1980 and 2013 the total number of adults with known diabetes more than doubled, rising from 1,601,063 to 3,394,039 (Table 2). The crude prevalence of diabetes in men increased from 3.3% in 1980 to 7.1% in 2013 (+115%) and from 4.7% to 6.8% in women (+45%). The prevalence did not increase linearly during the period: it remained almost stable during the eighties (around 3.5% in men and 4.5% in women), and then started to increase from the beginning of the nineties, reaching 7% in 2013 in both genders.

Diabetes prevalence increased in men and women, across all ages, educational levels, and areas of Italy, but with different patterns. The highest increase was among men aged 65 years and over (18% in 2013), and those with a lower educational level, while, for women, the prevalence increased more among those with a higher level of education compared to those with lower levels of education, as well as those in the oldest age group, but at a lower rate than among men, reflecting the smaller increase in prevalence shown by women.

In 2013, the standardised prevalence of diabetes was higher in men compared to women, in individuals with a low educational level compared to those with a high level of education and in Southern regions of Italy compared to

Table 1 Characteristics of Italian adults aged ≥ 20 years from 1980 to 2013[a].

Unweighted number	1980	1983	1987	1990	1994	2000	2005	2013
	53,078	64,920	56,874	50,853	48,110	111,885	103,250	97,182
Sex (%)								
Male	47.9	47.8	47.9	47.7	47.8	48.0	47.9	47.8
Female	52.1	52.2	52.1	52.3	52.2	52.0	52.1	52.2
Age (%)								
20–49	56.8	55.8	57.5	57.8	56.6	54.7	53.3	49.3
50–64	24.5	25.8	24.3	23.7	23.0	23.2	22.9	24.7
≥65	18.8	18.4	18.3	18.6	20.3	22.1	23.8	26.0
Educational level (%)								
High	17.5	20.7	23.4	26.9	29.5	33.8	36.8	44.2
Middle	19.8	22.4	29.3	32.6	33.2	33.1	34.7	34.2
Low	62.7	55.9	47.3	40.5	37.3	33.1	28.4	21.6
Missing	0	1.0	0	0	0	0	0	0
BMI (%)								
Underweight		3.2		2.7	3.4	3.3	3.2	3.0
Normal		56.9		52.0	56.3	54.3	52.1	51.4
Overweight		30.4		29.8	32.0	33.0	34.8	34.0
Obese		6.2		6.1	7.3	9.0	9.9	11.4
Missing		3.3		9.5	0.9	0.3	0	0.3
Area (%)								
North	46.8	47.1	46.3	46.2	46.3	46.1	46.3	46.0
Centre	19.9	19.7	19.6	19.7	19.8	19.7	19.6	20.0
South	33.3	33.2	34.1	34.1	34.0	34.3	34.2	34.1

[a] Estimates are weighted.

those in the North. However, these differences varied across the period. It is only since 2005 that prevalence in men exceeded that in women. Differences by educational level in men were absent at the beginning of the eighties, emerged during the nineties and increased during the last decade as a consequence of the steep increase in prevalence among those with a lower level of education; in women differences by education were evident from the start of the study period, showing a slightly declining gradient across the period because of the more marked increase among those with a high level of education compared to those with a lower educational level. Conversely, a North-South gradient was constant across all the surveys. One third of the increase in men and two thirds in women is due to the ageing of the population, as shown by the trend in age-standardised prevalence, that increased by 79% in men and 14% in women (compared to, respectively, 119% and 45% increase when crude prevalence is considered).

Overweight and obesity also increased both in men (more), and in women during the period, but to a lesser extent than diabetes. The highest increase occurred among people aged 65 or more: +43.5% in men and +28.5% in women (data not shown).

In non-adjusted logistic regression models (Table 3) the odds of prevalence increased, between 1983 and 2013, to 1.96 in men and 1.52 in women. Adjustment for age (Table 3 Model 2) reduced the association with the year of survey in both men (1.58) and women (1.16), resulting in a difference in the ORs of 39% in men and 69% in women. Inclusion of overweight and obesity in the model 3 further attenuated the OR both in men (1.46), and, to a lesser extent in women (1.11) (model 3 versus model 2: $p < 0.0001$ for both genders), the difference in the ORs being 21% in men, and 31% in women. The further inclusion of educational level resulted in no effect over OR (data not shown).

Discussion

Between 1980 and 2013 the number of people with known diabetes in Italy more than doubled, to stand at nearly 3.4 million in 2013, close to the number predicted by the International Diabetes Federation [1]. The secular trend in prevalence shows two different patterns in men and women. Prevalence in men has progressively increased, more steeply from the beginning of the nineties, across all age groups, educational levels and areas of Italy; the steepest increase was observed in men with a low educational level and in those aged 65 years and over; given the large number of men in the oldest age group, the steep increase in prevalence in this age group is the main contributor to the increase in the absolute number of men with diabetes. On the contrary, the prevalence increase among women was very slight, more noticeable in the oldest group, but its magnitude was far less than in men. As a consequence of these two patterns, differences by educational level emerged and gradually widened in men, while decreasing slightly in women. Similarly to that reported in other countries [7], the excess prevalence in women at the beginning of the eighties has progressively shifted to a higher prevalence in men. The different rates of increasing prevalence between men and women (i.e. steep in men and slow in women) are concordant with the current literature [7,11], but the rate was somewhat slower than that recorded in other countries such as Canada [4] or

Table 2 Secular trends of crude and standardised diabetes and of overweight/obesity prevalence (95% Confidence Intervals) in Italian men aged ≥20 years by age, area of residence and educational level.

Men	1980	1983	1987	1990	1994	2000	2005	2013	Variation 1980–2013%
Diabetes									
Number	626,651	783,767	677,508	667,389	883,300	1,030,934	1,209,863	1,647,353	+162
Crude Prevalence	3.3 (3.1–3.5)	4.0 (3.8–4.2)	3.4 (3.1–3.6)	3.2 (3.0–3.5)	4.2 (3.9–4.4)	4.7 (4.5–4.9)	5.4 (5.2–5.6)	7.1 (6.8–7.3)	+115
STD Prevalence	3.8 (3.4–4.0)	4.4 (4.1–4.6)	4.1 (3.8–4.4)	3.9 (3.6–4.2)	4.9 (4.6–5.2)	5.2 (5.0–5.4)	5.7 (5.5–5.9)	6.8 (6.6–7.0)	+79
Age									
20–49	1.1 (0.9–1.3)	1.2 (1.1–1.4)	0.8 (0.7–1.0)	0.8 (0.7–1.0)	0.9 (0.8–1.1)	1.0 (0.9–1.1)	1.0 (0.9–1.2)	1.2 (1.1–1.4)	+9
50–64	5.4 (4.8–6.0)	6.3 (5.8–6.8)	5.0 (4.5–5.5)	5.5 (4.9–6.1)	6.1 (5.5–6.7)	7.5 (7.0–7.9)	7.7 (7.2–8.2)	8.8 (8.3–9.3)	+63
≥ 65	7.7 (6.9–8.4)	10.0 (9.2–10.8)	10.3 (9.4–11.2)	9.1 (8.2–10.0)	12.6 (11.5–13.6)	12.5 (11.9–13.1)	14.6 (13.9–15.3)	18.1 (17.4–18.8)	+135
Education									
High-std	3.9 (3.1–4.8)	4.6 (3.9–5.4)	3.7 (2.9–4.7)	3.6 (2.9–4.6)	4.3 (3.5–5.2)	5.1 (4.5–5.6)	5.0 (4.6–5.5)	5.6 (3.3–6.1)	+44
Medium-std	5.7 (4.5–7.2)	4.9 (4.2–5.7)	4.6 (3.8–5.4)	4.4 (3.6–5.3)	5.0 (4.3–5.9)	4.9 (4.5–5.4)	5.8 (5.4–6.2)	7.2 (6.8–7.6)	+26
Low-std	3.6 (3.3–3.9)	4.4 (4.1–4.7)	4.1 (3.8–4.5)	4.4 (3.9–4.8)	5.4 (4.9–5.9)	5.8 (5.4–6.2)	6.7 (6.3–7.1)	8.5 (7.8–9.3)	+136
Area									
North-std	3.4 (3.0–3.8)	3.9 (3.6–4.3)	3.6 (3.2–4.1)	3.7 (3.3–4.1)	4.5 (4.0–4.9)	4.7 (4.5–5.0)	5.1 (4.8–5.4)	6.4 (6.1–6.7)	+88
Centre-std	3.9 (3.4–4.3)	5.0 (4.5–5.5)	4.3 (3.7–5.0)	4.1 (3.5–4.7)	5.1 (4.5–5.8)	5.1 (4.6–5.5)	5.8 (5.3–6.2)	6.7 (6.2–7.2)	+72
South-std	4.1 (3.6–4.7)	4.7 (4.2–5.2)	4.6 (4.2–5.1)	4.0 (3.6–4.5)	5.4 (4.9–6.0)	5.9 (5.6–6.2)	6.6 (6.2–6.9)	7.5 (7.2–7.9)	+83
Overweight/obesity									
STD Prevalence		43.6 (43.1–44.2)		45.0 (44.4–45.6)	49.0 (48.4–49.7)	51.6 (51.2–52.0)	53.7 (53.2–54.1)	53.7 (53.2–54.1)	+23
Women	**1980**	**1983**	**1987**	**1990**	**1994**	**2000**	**2005**	**2013**	**Variation 1980–2013%**
Diabetes									
Number	974,412	1,066,661	967,382	938,755	1,191,386	1,254,714	1,375,729	1,746,686	+79
Crude Prevalence	4.7 (4.4–4.9)	5.0 (4.7–5.2)	4.4 (4.2–4.6)	4.2 (3.9–4.4)	5.2 (4.9–5.4)	5.3 (5.1–5.5)	5.6 (5.4–5.8)	6.8 (6.6–7.1)	+45
STD Prevalence	5.0 (4.7–5.2)	5.2 (4.9–5.4)	4.7 (4.5–5.0)	4.5 (4.2–4.7)	5.3 (5.0–5.6)	5.1 (4.9–5.2)	5.2 (5.0–5.3)	5.8 (5.6–6.0)	+14
Age									
20–49	1.1 (1.0–1.3)	1.0 (0.9–1.2)	0.7 (0.5–0.8)	0.5 (0.4–0.7)	0.7 (0.6–0.8)	0.8 (0.7–0.9)	0.8 (0.7–0.9)	1.4 (1.2–1.5)	+27
50–64	6.1 (5.5–6.7)	6.9 (6.4–7.4)	5.5 (4.9–6.0)	5.7 (5.1–6.2)	6.8 (6.2–7.5)	5.9 (5.5–6.3)	6.1 (5.7–6.6)	5.9 (5.5–6.3)	- 3
≥65	12.6 (11.7–13.5)	13.0 (12.2–13.8)	13.3 (12.4–14.2)	11.9 (11.0–12.8)	14.2 (13.2–15.1)	14.0 (13.4–14.5)	14.5 (13.9–15.1)	16.7 (16.1–17.3)	+33
Education									
High-std	1.9 (1.4–2.7)	2.2 (1.7–2.9)	4.3 (3.2–5.7)	2.6 (1.8–3.8)	3.2 (2.5–4.1)	2.8 (2.4–3.3)	3.7 (3.3–4.2)	3.6 (3.2–3.9)	+89
Medium-std	3.6 (2.8–4.6)	3.1 (2.5–3.8)	3.9 (3.2–4.8)	2.9 (2.3–3.7)	3.6 (2.9–4.4)	3.5 (3.1–4.0)	4.4 (4.1–4.8)	5.8 (5.4–6.2)	+61
Low-std	5.2 (4.9–5.5)	5.7 (5.4–5.9)	4.9 (4.6–5.3)	5.0 (4.7–5.3)	6.2 (5.8–6.6)	5.9 (5.6–6.2)	6.0 (5.7–6.4)	7.8 (7.1–8.6)	+50
Area									
North-std	4.0 (3.6–4.4)	4.4 (4.1–4.7)	4.0 (3.6–4.3)	3.5 (3.2–3.9)	4.2 (3.9–4.6)	4.5 (4.3–4.8)	4.1 (3.8–4.3)	5.0 (4.7–5.3)	+25
Centre-std	5.0 (4.6–5.4)	4.8 (4.4–5.3)	4.8 (4.3–5.4)	4.5 (4.0–5.1)	5.2 (4.7–5.9)	4.7 (4.3–5.1)	5.3 (4.9–5.8)	5.6 (5.2–6.1)	+12
South-std	6.5 (5.9–7.1)	6.7 (6.2–7.3)	5.8 (5.3–6.4)	6.0 (5.5–6.5)	6.9 (6.4–7.5)	6.1 (5.9–6.4)	6.7 (6.4–7.0)	7.2 (6.9–7.5)	+10
Overweight/obesity									
STD Prevalence		30.5 (30.1–31.0)		29.1 (28.6–29.7)	31.7 (31.2–32.3)	33.3 (33.0–33.7)	35.2 (34.8–35.6)	34.4 (34.0–34.8)	+13

Table 3 Secular trends in diabetes prevalence among Italian adults aged ≥20 years; Odds Ratios and 95% Confidence Intervals.

	Model 1		Model 2		Model 3	
	OR	95% CI	OR	95% CI	OR	95% CI
Men						
1983	1		1		1	
1990	0.83	(0.76–0.91)	0.86	(0.78–0.94)	0.85	(0.77–0.93)
1994	1.05	(0.96–1.14)	1.07	(0.98–1.17)	1.01	(0.93–1.11)
2000	1.22	(1.14–1.31)	1.14	(1.06–1.23)	1.05	(0.98–1.13)
2005	1.45	(1.35–1.55)	1.30	(1.21–1.39)	1.17	(1.09–1.26)
2013	1.96	(1.83–2.09)	1.58	(1.47–1.69)	1.40	(1.31–1.51)
Women						
1983	1		1		1	
1990	0.81	(0.74–0.87)	0.84	(0.78–0.92)	0.83	(0.76–0.91)
1994	1.00	(0.93–1.08)	1.02	(0.95–1.11)	1.01	(0.93–1.09)
2000	1.11	(1.05–1.18)	0.99	(0.93–1.06)	0.94	(0.88–1.01)
2005	1.20	(1.13–1.27)	1.01	(0.95–1.08)	0.94	(0.88–1.01)
2013	1.52	(1.43–1.61)	1.16	(1.10–1.24)	1.09	(1.02–1.16)

Model 1: crude.
Model 2: age-adjusted.
Model 3: model 2 + BMI in two categories (Overweight and Obesity).

the USA [12]. The wide increase in the oldest age groups is very different from that reported in other countries, where prevalence increased across all age groups, and to a greater extent in the younger population [2,4].

As showed by the results of multivariate logistic regression models, in Italy, more than one third (39%) of the increased prevalence among men, and about two thirds (69%) of the increase in women, is due to the ageing of the population over the 34-year follow-up period. In the same period, prevalence of overweight and obesity contribute to the increase of prevalence of 21% in men and 31% in women. The non-linearity of the association between BMI and diabetes had already been demonstrated in the United States during the nineties [13]; indeed, it has been shown that while obesity explains most of the increase in diabetes among the youngest age-group, it only partly explains the increase in those aged 65 and over [2], and that ageing men have a higher predisposition to central fat deposition, that is associated (in men) with reduced oxidation of fat; in short, men are more prone than women to develop diabetes in response to obesity [11]. Indeed, the main increase in prevalence of overweight and obesity occurred in the eldest age groups, particularly in men. Unfortunately, waist circumference, which is now considered a better predictor of diabetes [14], was not recorded in the HIS.

The reasons for these different trends in diabetes prevalence are difficult to explain from cross-sectional data. Time trends in prevalence are necessarily related to changes in incidence and (or) mortality; however, there are no data available from the same surveys on trends in incidence of diabetes and survival of patients with known diabetes.

As for incidence, there are conflicting data from developed countries: a Canadian study reported an increase between 1995 and 2005 [4]; however, in Scotland incidence has stabilised in recent years [15], in Sweden there has been a modest decrease [16], and in the USA, a sharp increase between 1990 and 2008 was followed by a significant decline [17]. It has been argued that the observed stabilisation of incidence recently reported in several countries could be related to the increased awareness of the disease and more frequent diagnosis, that have led to a reduction in the number of cases of undiagnosed diabetes [4,16], as has been reported in Italy [18], and, recently, in Germany [19]. Despite the absence of data in Italy about the time trend in incidence at national level, there is no reason why the incidence pattern should be different from that found in other European countries.

Increased awareness of the disease could also have played an important role in increasing prevalence rates. In the early 1990s, the results of major trials [20] were published that demonstrated the relationship between hyperglycemia and complication occurrence. In addition, even if the prevalence of diabetes has risen almost linearly since 1990, changes in diagnostic criteria that were introduced in the late 1990s, when the WHO lowered the fasting plasma glucose threshold for diabetes from 7.8 to 7.0 mmol/L [21], could have affected the trend in prevalence. In Italy, in 1987, a specific diabetes Law defined diabetes as a “high-interest social disease” and established diabetes specialist services within a multidisciplinary/interdisciplinary departmental system. The establishment of diabetes centres all over Italy may have further increased awareness causing the identification of diabetes cases previously undiagnosed, whose prevalence has reduced in recent years (18).

The second driving force behind increased prevalence is the improvement in the survival rate of patients with established diabetes. Again, we do not have data at national level, but a decrease in mortality among patients with diabetes has been reported in several countries [4,22,23] and has likely also occurred in Italy, alongside improvements in the quality of diabetes care [18,24]; furthermore, several local Italian studies have shown that the increased mortality of patients with diabetes compared to those without the disease, is lower than that reported in other countries [6,25–27]. Consequently, we argue that the increasing prevalence of age-adjusted diabetes is more likely due to declining mortality than to increasing incidence, similarly to that reported among other European populations [15].

Moreover, increased awareness of the disease together with improvements in the care of people with known diabetes is likely to have had a greater effect on those who, traditionally, tend to disregard their health, and make less use of health care services, as is the case of men, the elderly and those with a low socioeconomic status. This would contribute to explain the steeper rise in prevalence of the disease among the oldest age group and men with a low level of education, and the overall increase in diabetes compared to women.

Our study has several limitations. First, we used a self-reported diagnosis of diabetes, the validity of which could be questioned; however the question about diabetes is largely used in health interview surveys and its validity

has been confirmed [13,28]. Similarly, the validity of reported height and weight (and, consequently, of BMI) could also be questioned; however, it has been reported that even if systematic errors in self-reported height and weight do exist, and that BMI is generally underestimated both in men and (more) in women, self-reported height and weight are valid measures for detecting associations between height and weight and disease in epidemiological studies [29]. Second, all cases of undiagnosed diabetes were excluded, thus resulting in an underestimate of the "true" prevalence of the disease. Third, we could not differentiate between type 1 and type 2 diabetes; however, as the latter accounts for more than 90% of cases, our results largely reflect the prevalence of type 2 diabetes. Fourth, the surveys did not include institutionalised people, among whom the prevalence of diabetes is unknown and could differ from that of the non-institutionalised population, thus affecting prevalence in the general population. On the other hand, the main strength of the study is that we used data from the health interview surveys conducted by the National Institute of Statistics on representative samples of the Italian population using the same methodology and the same set of questions for diabetes and for BMI over the last 34 years. There are very few published studies that cover such a long time-span describing the trend in diabetes at national level, and none from Southern Europe [3,7].

In conclusion, we have shown that in Italy, over a 34-year period, the number of people with diabetes and the prevalence of the disease have increased, without a parallel increase in overweight and obesity. Population ageing plays an important role in increasing the absolute number of people with diabetes; other than ageing, our data are not sufficient to fully explain the reason why prevalence is increasing, however we suggest that improved survival can be reasonably considered as a more significant contributor than changes in incidence. As the largest increase in the number of people affected by the disease is in the group of the population aged 65 and over, a group that is expected to grow, the number of people with diabetes is set to rise dramatically in the near future, and will represent a severe challenge for the national health system.

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Conflicts of interest

None.

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